



Python Programming Basics





PYTHON FUNDAMENTALS

In this chapter, you'll learn about:

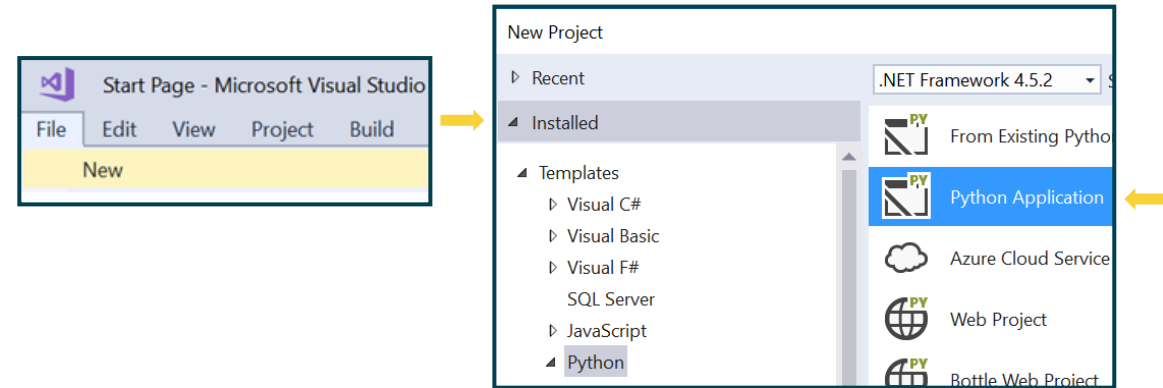
- Python in Visual Studio
- Basic statements
- Numbers, strings and Boolean variables
- Keyboard input
- Screen output
- Casting





WHY VISUAL STUDIO?

- VS is a superb development environment
 - Excellent debugging and testing facilities

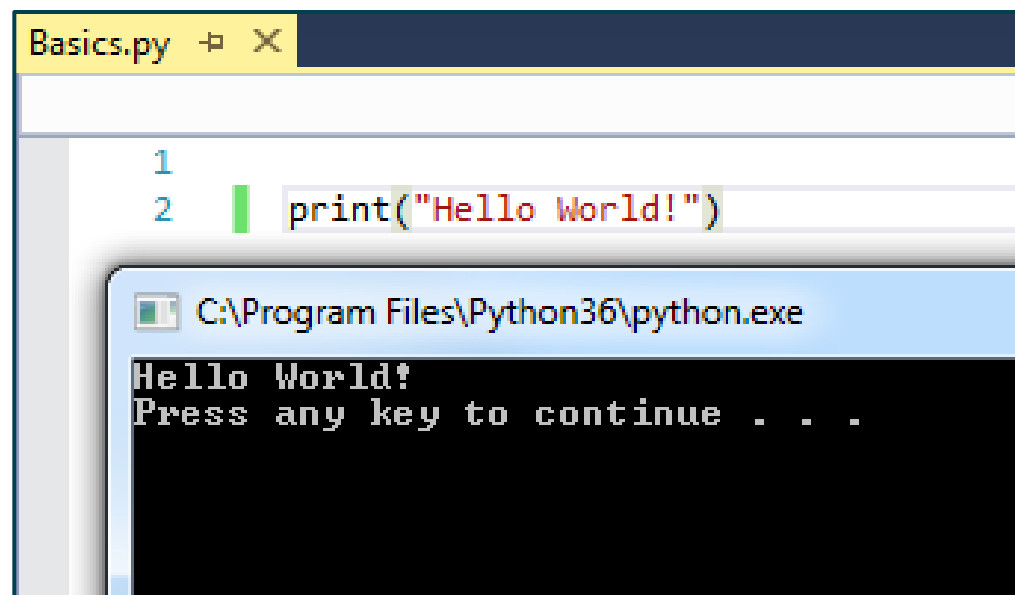


- There are many online compilers
 - <https://repl.it/languages/python3>



PRINTING TO CONSOLE WINDOW

- Print ('Hello World')
- Click  or **F5** key to run
- Press **Ctrl-F5** to run without debugging



The screenshot shows a Python IDE window titled 'Basics.py'. The code editor contains two lines of Python code: `1` and `2 print("Hello World!")`. A green cursor is positioned at the start of line 2. Below the code editor, a console window is open, displaying the output of the script. The console title bar reads 'C:\Program Files\Python36\python.exe'. The console output shows 'Hello World!' on the first line and 'Press any key to continue . . .' on the second line.

```
Basics.py  + X
1
2 print("Hello World!")

C:\Program Files\Python36\python.exe
Hello World!
Press any key to continue . . .
```



COMMENTS IN CODE

- Help readers understand your code
- Use a `#` to ignore the following chars
- Are ignored when code runs

```
# process user information  
print('Hello World!') # display greetings
```





DATA TYPES IN PYTHON

There Are Three Basic Variable Types:

- Numbers: Integer and Float
 - 1,2,3, 1.23, 0.0005
- Character or String
 - 'Hello world' or "Hello world"
- Boolean **True** or **False**
(case-sensitive)

```
age=21  
salary = 2000.78  
companyName='QA Ltd'  
isRegistered = True  
hasLicence = False
```

type is determined
automatically;
value can change

VARIABLE NAMING STANDARDS



- Use letters, not punctuation

salay\$ = 1500 ❌

my-city='London' ❌

my city='London' ❌

my_city='London' ✔️

myCity='London' ✔️

- No 'reserved word'

print = 10 ❌

VARIABLE NAMING STANDARDS

Case Sensitive

- Use lowercase letters for consistency

What will be
displayed?

```
age=32  
Age=21  
print(age)
```

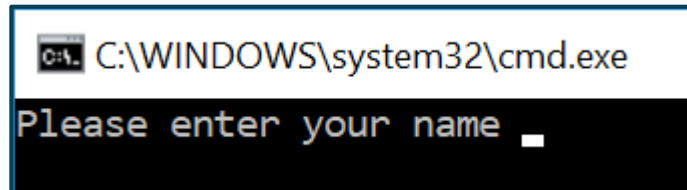
32



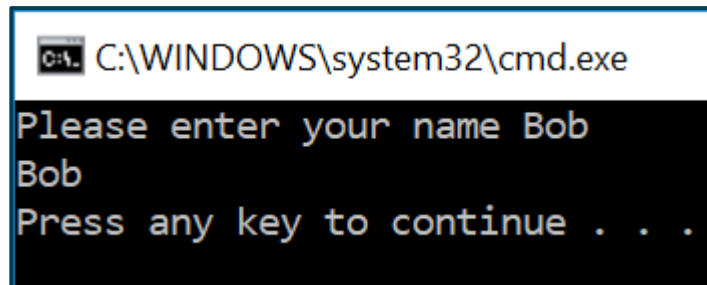
USER INPUT USING KEYBOARD

Input with a Prompt `input(<prompt>)`

```
name = input('Please enter your name ')\nprint(name)
```

A screenshot of a Windows command prompt window. The title bar shows 'C:\WINDOWS\system32\cmd.exe'. The command prompt displays the text 'Please enter your name ' followed by a cursor (a small white square) on the same line.

```
C:\WINDOWS\system32\cmd.exe\nPlease enter your name _
```

A screenshot of a Windows command prompt window. The title bar shows 'C:\WINDOWS\system32\cmd.exe'. The command prompt displays the text 'Please enter your name Bob' on one line, 'Bob' on the next line, and 'Press any key to continue . . .' on the third line.

```
C:\WINDOWS\system32\cmd.exe\nPlease enter your name Bob\nBob\nPress any key to continue . . .
```



PUTTING STRINGS TOGETHER

```
username = 'Bob'  
print('Hello' , username)  
print('Hello' + username)  
print('Hello ' + username)
```

```
Hello Bob  
HelloBob  
Hello Bob
```

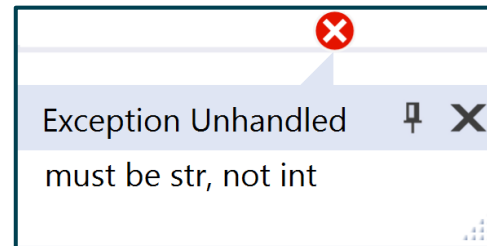
Cannot add numbers and strings

```
age = 21  
print('Your age is ' + age)  
Message = 'Your age is ' + age
```



keyboard input is always text...
even if it 'looks' like a number

```
age = input('Please enter your age ' )  
age = age + 1
```



CASTING

Keyboard Is Text, So We Use Casting to Convert It to Other Types

```
age = int(input('What is your age? '))  
age = age + 1  
print('Next year you will be', age, 'years old')
```



What is your age? 21
Next year you will be 22 years old

```
age = input('What is your age? ')  
age = int(age)  
age = age + 1  
print('Next year you will be', age, 'years old')
```





CASTING A NUMBER TO STRING

Use the `str()` function

```
## Find the average of a few numbers
```

```
total = 1 + 3 + 5 + 7 + 9 + 11  
average = total / 6
```

```
print("Total is = " + str(total))  
print("Average is = " +  
      str(average))
```





CASTING FLOATS

```
price = int(input('What is the price? '))  
totalPrice = price * 1.2
```



```
price = float(input('What is the price? '))  
totalPrice = price * 1.2
```





Guided Task

- In this task, you will write code to:
 - Edit code in Visual Studio
 - Compile and run your code
 - Input values into variables
 - Lab duration: 30 minutes
- Please open the instructions in cloud academy

SUMMARY

In this chapter you learned about:

- Python in Visual Studio
- Basic statements
- Numbers, strings and Boolean variables
- Keyboard input
- Screen output
- Casting





FURTHER READING

- <https://www.python.org/>
- <https://www.python.org/dev/peps/pep-0008/#a-foolish-consistency-is-the-hobgoblin-of-little-minds>





Thank you!

Hope you enjoyed this learning journey